

Supplementary Material for “DisCo-CFL: Calibrated, Fair and Explainable Clustered Federated Learning via Disattenuated Class-Conditional Gradient Signatures”

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This supplement contains the full definitions and statements (Section S1), all proofs (Section S2), the Lean 4 formalisation (Section S3), and additional experimental details and results (Section S4). Code, raw results and the Lean project are available at https://github.com/vinhqdang/cluster_federated_learning. Results carry the prefix “S”. They correspond to the main paper as follows: Proposition S1 is Proposition 1; Lemma S1 and Theorem S1 together form Theorem 1; Theorems S2 and S3 are Theorems 2 and 3; Corollary S1 is Corollary 1; Proposition S2 is Proposition 2; Propositions S3 and S4 together form Proposition 3.

S1 Definitions and full statements

Definition S1 (Concept equivalence and conflicts). Clients i and j *conflict on class* $c \in S_i \cap S_j$ if $\mathcal{P}_i(X | Y = c) \neq \mathcal{P}_j(X | Y = c)$. They are *compatible* if they conflict on no shared class. Clients belong to K latent concept groups $G_1, \dots, G_K$: $\mathcal{P}_i(X | Y = c) = \mathcal{P}^{(k)}(X | Y = c)$ for $i \in G_k$.

S1.1 Method definitions

S1.1.1 Class-conditional gradient signatures (client side)

After T_0 warm-up rounds of FedAvg the server broadcasts a reference model w and the seed of a count-sketch $\Phi : \mathbb{R}^d \rightarrow \mathbb{R}^p$ [1, 2] ($p = 1024$). For a sample (x, c) let $g(x, c) = \Phi \nabla_w \ell(w; x, c) \in \mathbb{R}^p$. For every class $c \in S_i$ the client randomly splits its class- c samples into halves A_i^c, B_i^c of size h_i^c and sends

$$a_i^c = \frac{1}{h_i^c} \sum_{s \in A_i^c} g(x_s, c), \quad b_i^c = \frac{1}{h_i^c} \sum_{s \in B_i^c} g(x_s, c).$$

Per-sample sketched gradients are clipped to a shared norm C_{clip} (0.5 in all experiments) before averaging. Per-sample gradients at a trained reference model are extremely heavy-tailed: in our Fashion-MNIST runs the median norm is 0.08 and the 99th percentile about 23, so a handful of hard examples would otherwise dominate a class signature. The expected clipped gradient is still a functional of $\mathcal{P}_i(X | Y = c)$, so invariance is preserved, and clipping makes per-sample signatures bounded, hence sub-Gaussian as required by Assumption S1. The upload happens *once* and costs $2p|S_i| \leq 2pC$ floats, independent of the model size d . For our small CNN ($d \approx 4.7 \times 10^4$) this is comparable to one model upload; for realistic models with $d \gg 2pC$ it is negligible. For differential privacy (Proposition S2), Gaussian noise is added to each clipped half-mean.

Algorithm S1 DisCo-CFL

- 1: Warm-up: T_0 rounds of FedAvg on all clients $\rightarrow$ reference model w ; broadcast w and sketch seed
 - 2: **for** each client i **in parallel** **do** $\triangleright$ once
 - 3: **for** $c \in S_i$ **do** split class- c samples into halves, send (a_i^c, b_i^c) (per-sample clipped; noised if DP)
 - 4: **end for**
 - 5: **end for**
 - 6: Server: R_i^c by (S1); $\hat{\rho}_{ij}^c, W_{ij}^c$ by (S2)
 - 7: Clusters $\leftarrow$ singletons of clients with enough evidence
 - 8: **while** $\exists A, B$ with $L(A, B) \geq \tau$ **do** merge the eligible pair with the largest $\sum_c \text{den}_{AB}^c$; log it
 - 9: **end while**
 - 10: Attach low-evidence clients and non-significant singletons; flag anomalies
 - 11: Output clusters, certificates, explanations, audit log; run FedAvg within each cluster from w
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S1.1.2 Disattenuated similarity (server side)

Let $m_i^c = (a_i^c + b_i^c)/2$. The split-half reliability and its Spearman–Brown extension to the full mean are

$$r_i^c = \cos(a_i^c, b_i^c), \quad R_i^c = \frac{2r_i^c}{1 + r_i^c} \quad (R_i^c := 0 \text{ if } r_i^c \leq 0). \quad (\text{S1})$$

The calibrated similarity of i and j on class c is

$$\hat{\rho}_{ij}^c = \text{clip}_{[-1,1]} \left(\frac{\cos(m_i^c, m_j^c)}{\sqrt{R_i^c R_j^c}} \right), \quad W_{ij}^c = R_i^c R_j^c, \quad (\text{S2})$$

where W_{ij}^c is the evidence weight. In the population version of (S2) the attenuation factors cancel exactly (Lemma S1), so $\hat{\rho}$ targets $\rho_{ij}^c = \cos(\mu_i^c, \mu_j^c)$, the cosine between expected class-conditional gradients, whatever the sample sizes and noise levels.

S1.1.3 Threshold clustering with bottleneck linkage

For two sets of clients A, B and a class c , let $\text{num}_{AB}^c = \sum_{i \in A, j \in B} W_{ij}^c \hat{\rho}_{ij}^c$ and $\text{den}_{AB}^c = \sum W_{ij}^c$. The linkage is

$$L(A, B) = \min_c \frac{\text{num}_{AB}^c + \lambda}{\text{den}_{AB}^c + \lambda} \quad \text{if } \sum_c \text{den}_{AB}^c \geq e_{\min}, \text{ else } -\infty. \quad (\text{S3})$$

The *minimum* over classes encodes the definition of concept equivalence: two groups share a concept only if every class-conditional gradient agrees, so a label swap affecting two of ten classes is not averaged away. The shrinkage λ pulls classes with little evidence towards 1 (“no evidence of a difference”). Starting from singletons, the server repeatedly merges, among the pairs with $L \geq \tau$, the pair with the *largest shared evidence*. Well-supported groups therefore form first, and clients that carry little information on the discriminative classes join them afterwards instead of bridging different groups. Merging stops when no pair reaches τ . Because the scale is calibrated, τ has a fixed meaning: we use $\tau = 0.8$ in *all* experiments, with $\lambda = 0.5$. Clients with total reliability $\sum_c R_i^c < 0.3$ are *low-evidence*: they are excluded from shaping clusters and attached to their most similar cluster afterwards. A singleton cluster is attached to its closest cluster unless the evidence *significantly* separates them (upper confidence bound below τ), in which case it is kept and flagged as *anomalous*. Each cluster then runs FedAvg from w . Algorithm S1 summarises the procedure.

S1.1.4 Inference: prior-corrected cluster models

Clusters are built so that their members share class-conditionals but may have very different label proportions. By Bayes' rule, the posterior of client i is obtained from that of its cluster k by re-weighting the prior (Proposition S4). Client i therefore predicts with

$$\hat{y}(x) = \arg \max_y [f_k(x)_y + \log \hat{p}_i(y) - \log \hat{p}_k(y)], \quad (\text{S4})$$

where f_k are the logits of the cluster model, $\hat{p}_i$ is the client's own (smoothed) label distribution, and $\hat{p}_k$ is the label distribution of the cluster's training data. The latter is a sum of label counts that the server can obtain by secure aggregation, so no individual histogram is revealed. This correction [3, 4] costs nothing at training time. It is exactly what concept-level clustering calls for: label skew is handled by the prior, not by more clusters. In the experiments we apply the same correction to *every* method, for a fair comparison.

S1.1.5 Transparency and accountability outputs

- **Class-level explanations.** For every pair of clusters the server reports the pooled per-class similarity and the set of *divergent classes* $\hat{D}_{AB} = \{c : \rho_{AB}^c < \tau\}$.
- **Heterogeneity diagnosis.** If $\hat{D}_{AB}$ contains every shared class, the server reports a global (feature-level) concept shift; if it is a strict subset, a class-specific shift on the listed classes; if $\hat{K} = 1$, it reports that the heterogeneity is explained by target and quantity skew only.
- **Assignment certificates.** Each client receives its bottleneck similarity to its own cluster and to the closest other cluster, the bottleneck classes, and standard errors from the spread over member pairs. A certificate is *confident* if $s_{\text{own}} - z \text{se} \geq \tau$ and $s_{\text{other}} + z \text{se} < \tau$ ($z = 2$); otherwise it is *uncertain* or *low-evidence*.
- **Audit log** of every merge (members, linkage, bottleneck class, evidence) and of every attachment or anomaly decision, so that each grouping can be traced and contested.
- **Newcomers** compute their signature at the same frozen w and join the cluster with the highest bottleneck similarity if it reaches τ . Otherwise they are reported as a *novel concept*.

S1.2 Full statements

Proofs are in Section S2; their deterministic parts are formalised in Lean 4 (Section S3). Write $\mu_i^c = \mathbb{E}_{x \sim \mathcal{P}_i(X|Y=c)}[g(x, c)]$ and $a_i^c = \|\mu_i^c\|^2$.

Proposition S1 (Invariance). μ_i^c is a functional of $\mathcal{P}_i(X | Y = c)$ only. Consequently, for compatible clients $\rho_{ij}^c = 1$ on every shared class, whatever their label proportions $\mathcal{P}_i(Y), \mathcal{P}_j(Y)$ and sample sizes n_i, n_j .

Assumption S1 (Noise). Per-sample signatures are $g = \mu_i^c + \Sigma^{1/2}z$ with independent z having independent, zero-mean, unit-variance, K_0 -sub-Gaussian coordinates. Let $\sigma^2 = \text{tr } \Sigma$, and define the effective ranks $r_s = \text{tr } \Sigma / \|\Sigma\|_{\text{op}}$ and $r_e = (\text{tr } \Sigma)^2 / \|\Sigma\|_F^2$.

Assumption S2 (Identifiability). If i, j conflict on c , then $\rho_{ij}^c \leq 1 - \Delta$ for some $\Delta \in (0, 1]$.

For a pair and a class let $h = \min(h_i^c, h_j^c)$, $a = \min(a_i^c, a_j^c)$, and let $\kappa = \sigma^2 / (ah)$ be the inverse signal-to-noise ratio of a half-mean.

Lemma S1 (Exact calibration at the population level). *Replacing each inner product and squared norm in (S1)–(S2) by its expectation gives exactly ρ_{ij}^c .*

Theorem S1 (Concentration). *Under Assumption S1, let $\kappa \leq 1$ and $\delta \in (0, 1)$. There are constants c_0, C_1 depending only on K_0 such that, if*

$$\eta := \left(\sqrt{\kappa/r_s} + \kappa/\sqrt{r_e} \right) \sqrt{\log(12/\delta)} \leq c_0,$$

then with probability at least $1 - \delta$

$$|\hat{\rho}_{ij}^c - \rho_{ij}^c| \leq C_1 \eta, \quad |R_i^c - R_i^{c,\star}| \leq C_1 \eta, \quad R_i^{c,\star} := \frac{a_i^c}{a_i^c + \sigma^2/(2h_i^c)}.$$

In contrast, the raw cosine $\cos(m_i^c, m_j^c)$ converges to $\rho_{ij}^c \sqrt{R_i^{c,\star} R_j^{c,\star}}$. It is biased towards 0 by a factor that depends on the sample sizes.

Theorem S2 (Safety, maximality, recovery). *Let $\gamma = \min(1-\tau, \tau-1+\Delta) > 0$ and $w_0 := 2\lambda(1-\tau)/\gamma$. Suppose that all estimates satisfy $|\hat{\rho}_{ij}^c - \rho_{ij}^c| \leq \gamma/2$ (event $\mathcal{E}$) and that every pair conflicting on a class c has evidence $W_{ij}^c > w_0$. Then the merging phase of Algorithm S1, in any merge order, satisfies:*

- (a) Safety: *no cluster ever contains two conflicting clients.*
- (b) Maximality: *any two output clusters with shared evidence $\geq e_{\min}$ contain a conflicting pair, so no two output clusters can be merged without mixing concepts.*
- (c) Exact recovery: *if every client holds every class on which some groups differ, and every pair of clients shares evidence $\geq e_{\min}$, the output partition equals $\{G_1, \dots, G_K\}$; in particular $\hat{K} = K$.*

Corollary S1 (Sample complexity and minority fairness). *Under Assumptions S1–S2, let $\gamma \leq 1/3$ and $\lambda < \gamma/(8(1-\tau))$. The conclusions of Theorem S2 hold with probability at least $1 - \delta$ as soon as every reported half size satisfies*

$$h \geq h^\star := C_2 \frac{\sigma^2}{a} \max \left\{ \frac{\mathsf{L}}{r_s \gamma^2}, \frac{\sqrt{\mathsf{L}}}{\sqrt{r_e} \gamma}, 1 \right\}, \quad \mathsf{L} := \log \frac{12N^2 C}{\delta}. \quad (\text{S5})$$

$h^\star$ does not depend on the sizes of the groups $|G_k|$, on the number of clients per group, or on the other clients' dataset sizes (apart from $\log N$). A concept group made of a single client is recovered under the same condition as a group of a hundred.

Theorem S3 (Minimax lower bound). *Let $\xi \sim \mathcal{N}(0, \Sigma)$ and $\|\mu\|^2 = a$. For any test that decides from n per-sample signatures of client j , even when μ_i is known exactly, whether $\rho_{ij} = 1$ or $\rho_{ij} = 1 - \Delta$, the worst-case error probability is at least*

$$\frac{1}{4} \exp(-n a \Delta r_s / \sigma^2).$$

Error at most δ therefore requires

$$n \geq \frac{\sigma^2}{a r_s \Delta} \log \frac{1}{4\delta}.$$

With τ in the middle of the admissible interval ($\gamma = \Delta/2$), the leading term of (S5) is

$$O\left(\frac{\sigma^2 \log(N^2 C / \delta)}{a r_s \Delta^2}\right).$$

The upper and lower bounds therefore agree in their dependence on noise σ^2 , signal a and effective rank r_s , and differ by a factor $1/\Delta$ and a $\log N$ term.

Proposition S2 (Differential privacy and its cost). *Suppose each released half-mean is*

$$\frac{1}{h} \sum_s \text{clip}_{C_{\text{clip}}}(g_s) + \xi_{\text{dp}}, \quad \xi_{\text{dp}} \sim \mathcal{N}(0, (2\sigma_{\text{dp}}C_{\text{clip}}/h)^2 I_p).$$

(i) *Every sample enters exactly one released vector, whose replace-one sensitivity is $2C_{\text{clip}}/h$. The signature is therefore (ε, δ) -DP with respect to replacing one record by another of the same label, with (ε, δ) given by the analytic Gaussian mechanism [5] at noise multiplier σ_{dp} (parallel composition). The class support S_i is released in the clear. (ii) The DP noise is independent of sampling noise. Lemma S1 therefore still holds, and Theorem S1 applies with the half-mean covariance $\Sigma/h + (2\sigma_{\text{dp}}C_{\text{clip}}/h)^2 I_p$. The recovery guarantee then requires, in addition to (S5),*

$$h \gtrsim \sigma_{\text{dp}} C_{\text{clip}} \left(\frac{\sqrt{\mathbb{L}}}{\sqrt{a} \gamma} + \frac{p^{1/4} \mathbb{L}^{1/4}}{\sqrt{a} \gamma} \right),$$

so the per-class sample size grows linearly in $\sigma_{\text{dp}} \propto 1/\varepsilon$. (iii) Without disattenuation, the raw cosine tends to $\rho \sqrt{R_i R_j}$ with

$$R \approx \frac{a}{a + 2p \sigma_{\text{dp}}^2 C_{\text{clip}}^2 / h^2}.$$

Unless $h \gg \sqrt{p} \sigma_{\text{dp}} C_{\text{clip}} / \sqrt{a}$, same-concept clients look unrelated.

Proposition S3 (Explanation fidelity). *On $\mathcal{E}$, if two output clusters A, B have pooled evidence $\text{den}_{AB}^c > w_0$ on a class c , then $c \in \hat{D}_{AB}$ if and only if $\mathcal{P}_A(X | Y = c) \neq \mathcal{P}_B(X | Y = c)$. The reported divergent classes are exactly the classes on which the two concepts differ.*

Proposition S4 (Prior correction is Bayes-optimal within a concept cluster). *Let cluster k consist of concept-equivalent clients with pooled distribution $\mathcal{P}_k(X, Y) = \mathcal{P}_k(Y) \mathcal{P}(X | Y)$, and let $q_k(y | x)$ be its Bayes posterior. For every member i , $\mathcal{P}_i(y | x) \propto q_k(y | x) \mathcal{P}_i(y) / \mathcal{P}_k(y)$. Hence (S4) with the true priors and $f_k = \log q_k$ is the Bayes-optimal classifier of client i . No clustering finer than concept equivalence can reduce the Bayes risk of any member.*

Remark 1 (Scope of the guarantees). Theorem S2 covers the merging phase. The attachment of low-evidence clients and of non-significant singletons is a fairness heuristic: it avoids training a client alone because of noise, may violate safety when evidence is weak, and is logged in the audit trail. Corollary S1 asks for a small λ so that a single pair of clients carries enough evidence. Our default $\lambda = 0.5$ is larger. It makes the method more conservative about splitting on thin evidence, and merges quickly happen between clusters whose pooled evidence far exceeds w_0 . Table S9 reports the sensitivity to λ , including values that satisfy the corollary.

S2 Proofs

Throughout, $C, C_1, C_2, \dots$ and c_0 denote constants that depend only on the sub-Gaussian constant K_0 and may change from line to line. We fix a class c and drop it from the notation when no confusion arises. Statements whose deterministic core is machine-checked in Lean 4 are marked with the name of the corresponding Lean theorem (Section S3).

S2.1 Proof of Proposition S1

The reference model w , the sketch Φ and the label c are the same for all clients, so

$$\mu_i^c = \int \Phi \nabla_w \ell(w; x, c) d\mathcal{P}_i(x | Y = c)$$

is determined by $\mathcal{P}_i(X | Y = c)$ alone. Neither the proportion $\mathcal{P}_i(Y = c)$ nor the number of samples enters. If i and j are compatible, then $\mathcal{P}_i(X | Y = c) = \mathcal{P}_j(X | Y = c)$ on each shared class, hence $\mu_i^c = \mu_j^c$ and $\rho_{ij}^c = 1$. Per-sample clipping is applied identically by all clients, so the same holds for the expected clipped signature. $\square$

S2.2 Proof of Lemma S1

Write the half-means as $a_i = \mu_i + \zeta_{i1}$ and $b_i = \mu_i + \zeta_{i2}$, where ζ_{i1}, ζ_{i2} are independent, zero-mean and have covariance Σ_i/h_i (any covariance works, and it may differ across clients). Let $t_i = \text{tr } \Sigma_i/h_i$. Then

$$\mathbb{E}\langle a_i, b_i \rangle = a_i, \quad \mathbb{E}\|a_i\|^2 = \mathbb{E}\|b_i\|^2 = a_i + t_i,$$

so the population split-half reliability is $r_i^* = a_i/(a_i + t_i)$ and

$$R_i^* = \frac{2r_i^*}{1 + r_i^*} = \frac{a_i}{a_i + t_i/2}.$$

For the full means, $\mathbb{E}\|m_i\|^2 = a_i + t_i/2 = a_i/R_i^*$, and by independence across clients $\mathbb{E}\langle m_i, m_j \rangle = \langle \mu_i, \mu_j \rangle$. The population cosine is therefore

$$\frac{\langle \mu_i, \mu_j \rangle}{\sqrt{(a_i/R_i^*)(a_j/R_j^*)}} = \frac{\langle \mu_i, \mu_j \rangle}{\sqrt{a_i a_j}} \sqrt{R_i^* R_j^*} = \rho_{ij} \sqrt{R_i^* R_j^*},$$

and dividing by $\sqrt{R_i^* R_j^*}$ gives ρ_{ij} exactly. (Lean: `spearman_brown, population_calibration`.) $\square$

S2.3 Proof of Theorem S1

Noise representation. A half-mean noise vector is $\zeta = (\Sigma/h)^{1/2} \bar{z}$ with $\bar{z} = h^{-1/2} \sum_s z_s$. Its coordinates are independent, zero-mean, unit-variance and CK_0 -sub-Gaussian.

Statistics. $\hat{\rho}_{ij}$ and R_i are functions of nine statistics:

$$X_1 = \langle a_i, b_i \rangle, \quad X_2 = \|a_i\|^2, \quad X_3 = \|b_i\|^2, \quad X_4 = \|m_i\|^2,$$

the same four statistics $X_5, \dots, X_8$ for client j , and $X_9 = \langle m_i, m_j \rangle$. We normalise them as $x_k = X_k/a_i$ for the statistics of i , $x_k = X_k/a_j$ for those of j , and $x_9 = X_9/\sqrt{a_i a_j}$. With $\kappa_i = \text{tr } \Sigma/(a_i h_i) \leq \kappa$, their expectations are

$$x^* = (1, 1 + \kappa_i, 1 + \kappa_i, 1 + \kappa_i/2, \dots, \rho_{ij}).$$

Each statistic is a constant, plus linear terms $\langle \mu, \zeta \rangle$, plus quadratic or bilinear forms in the ζ 's.

Linear terms. $\langle \mu_i, \zeta \rangle$ is sub-Gaussian with variance proxy

$$CK_0^2 \frac{\mu_i^\top \Sigma \mu_i}{h} \leq CK_0^2 \frac{a_i \|\Sigma\|_{\text{op}}}{h}.$$

After normalisation by a_i (or by $\sqrt{a_i a_j}$ for cross terms), it is, with probability $1 - \delta'$, at most

$$C \sqrt{\frac{\|\Sigma\|_{\text{op}}}{ah}} \sqrt{\log(2/\delta')} = C \sqrt{\kappa/r_s} \sqrt{\log(2/\delta')},$$

because $\|\Sigma\|_{\text{op}} = \sigma^2/r_s$.

Quadratic and bilinear terms. $\|\zeta\|^2 - \text{tr}(\Sigma/h)$ and $\langle \zeta, \zeta' \rangle$ for independent ζ, ζ' are centred quadratic forms $\bar{Z}^\top M \bar{Z}$ of the concatenated vector $\bar{Z}$, with $\|M\|_F \leq \|\Sigma\|_F/h$ and $\|M\|_{\text{op}} \leq \|\Sigma\|_{\text{op}}/h$. The Hanson–Wright inequality [6] bounds the deviation, with probability $1 - \delta'$, by

$$CK_0^2 \left(\|M\|_F \sqrt{\log(2/\delta')} + \|M\|_{\text{op}} \log(2/\delta') \right).$$

Since $\|\Sigma\|_F = \sigma^2/\sqrt{r_e}$, after normalisation by a this is at most

$$C \frac{\kappa}{\sqrt{r_e}} \sqrt{\log(2/\delta')} + C \frac{\kappa}{r_s} \log(2/\delta'),$$

and the second term is at most $C\eta^2 \leq Cc_0\eta$.

Union bound. Taking $\delta' = \delta/6$ over the at most six independent noise terms that appear in each group of statistics, we obtain $\|x - x^*\|_\infty \leq C\eta$ with probability $1 - \delta$.

Smoothness. Write $\hat{\rho} = F(x)$, where, before clipping,

$$r_i = \frac{x_1}{\sqrt{x_2 x_3}}, \quad R_i = \frac{2r_i}{1 + r_i} \quad (\text{similarly for } j), \quad \hat{\rho} = \frac{x_9}{\sqrt{x_4 x_8 R_i R_j}}.$$

By Lemma S1, $F(x^*) = \rho_{ij}$. For $\kappa \leq 1$ and $\|x - x^*\|_\infty \leq Cc_0$ with c_0 small enough, the argument stays in a compact set on which $x_2, x_3, x_4 \in [1/2, 3]$, $r_i \geq 1/5$ and $R_i \geq 1/3$. On that set F is continuously differentiable with gradient bounded by an absolute constant. Hence $|F(x) - F(x^*)| \leq C_1\eta$, and the same argument gives the bound for R_i . Clipping to $[-1, 1]$ can only reduce the error, because $\rho_{ij} \in [-1, 1]$ (Lean: `clip_reduces_error`). For the raw cosine,

$$\frac{x_9}{\sqrt{x_4 x_8}} \longrightarrow \frac{\rho_{ij}}{\sqrt{(1/R_i^*)(1/R_j^*)}} = \rho_{ij} \sqrt{R_i^* R_j^*}. \quad \square$$

S2.4 Proof of Theorem S2

We use two facts, both on $\mathcal{E}$.

Claim 1 (conflicting clusters cannot be linked). Let A, B be conflict-free and suppose some $i \in A$ and $j \in B$ conflict on class c . Since A is conflict-free, all members of A holding c share one class-conditional $\mathcal{P}_A^c$; likewise for B , and $\mathcal{P}_A^c \neq \mathcal{P}_B^c$. Hence *every* pair in $A \times B$ holding c conflicts on c . By Assumption S2 and $\mathcal{E}$, each such pair has $\hat{\rho} \leq 1 - \Delta + \gamma/2$, so

$$\text{num}_{AB}^c \leq (1 - \Delta + \gamma/2) \text{den}_{AB}^c.$$

Moreover

$$\frac{\text{num}_{AB}^c + \lambda}{\text{den}_{AB}^c + \lambda} < \tau \iff \text{den}_{AB}^c (\tau - 1 + \Delta - \gamma/2) > \lambda(1 - \tau).$$

Since $\tau - 1 + \Delta \geq \gamma$, the left side is at least

$$\text{den}_{AB}^c \frac{\gamma}{2} \geq W_{ij}^c \frac{\gamma}{2} > w_0 \frac{\gamma}{2} = \lambda(1 - \tau).$$

The minimum over classes in (S3) is therefore below τ . (Lean: `claim1_conflict_not_linked`.)

Claim 2 (compatible clusters are always linkable). If $A \cup B$ is conflict-free, then every pair in $A \times B$ that holds a class c has $\rho^c = 1$ (Proposition S1), so $\hat{\rho}^c \geq 1 - \gamma/2$. Every class term of (S3) is then at least

$$1 - \frac{\gamma}{2} \geq 1 - \frac{1 - \tau}{2} = \frac{1 + \tau}{2} > \tau,$$

and classes without evidence contribute 1. So $L(A, B) > \tau$ whenever the shared evidence is at least $e_{\min}$. (Lean: `claim2_compatible_linked`.)

Conclusion. (a) Singletons are conflict-free. By Claim 1, a merge is only performed between clusters whose union is conflict-free, so the invariant is preserved by every merge, in any order. (b) At termination, every pair of clusters with shared evidence $\geq e_{\min}$ has $L < \tau$; by Claim 2 their union is not conflict-free. (c) Under full support, two clients of different groups $k \neq l$ both hold a class on which $\mathcal{P}^{(k)}$ and $\mathcal{P}^{(l)}$ differ, so they conflict. By (a), each output cluster is contained in a single group. Two output clusters inside the same group have a conflict-free union and share evidence $\geq e_{\min}$, which contradicts (b). Each group thus forms exactly one cluster. (Lean: `merge_preserves_conflict_free`, `safety_all_merges`, `exact_recovery`.) $\square$

S2.5 Proof of Corollary S1

There are at most $N^2C/2$ pairwise class estimates and NC reliabilities; we apply Theorem S1 to each with $\delta/(N^2C)$. With $L = \log(12N^2C/\delta)$, the requirement $C_1\eta \leq \gamma/2$ holds if

$$\sqrt{\kappa/r_s} \sqrt{L} \leq \frac{\gamma}{4C_1} \quad \text{and} \quad \frac{\kappa}{\sqrt{r_e}} \sqrt{L} \leq \frac{\gamma}{4C_1}.$$

With $\kappa = \sigma^2/(ah)$ these are the first two terms of (S5) (Lean: `sample_size_sufficient`). The side conditions $\kappa \leq 1$ and $\eta \leq c_0$ are absorbed into C_2 and the third term. For the evidence condition: on the event, $R \geq R^* - \gamma/2 \geq 2/3 - \gamma/2$ because $\kappa \leq 1$. For $\gamma \leq 1/3$ this gives $W \geq 1/4$, so $W > w_0$ holds whenever $\lambda < \gamma/(8(1 - \tau))$ (Lean: `evidence_condition`). None of these conditions involves $|G_k|$. $\square$

S2.6 Proof of Theorem S3

Let e_1 be a top eigenvector of Σ ($\Sigma e_1 = \|\Sigma\|_{\text{op}} e_1$) and let $e_2 \perp e_1$ be a unit vector. With $s^2 = \Delta/2$ put

$$u = \sqrt{a}(\sqrt{1 - s^2} e_2 + s e_1), \quad v = \sqrt{a}(\sqrt{1 - s^2} e_2 - s e_1).$$

Then

$$\|u\|^2 = \|v\|^2 = a, \quad \cos(u, v) = 1 - 2s^2 = 1 - \Delta, \quad u - v = 2\sqrt{a} s e_1$$

(Lean: `lower_bound_construction`). Under H_0 the n samples of client j are i.i.d. $\mathcal{N}(u, \Sigma)$; under H_1 they are $\mathcal{N}(v, \Sigma)$; $\mu_i = u$ is known. Then

$$\text{KL}(\mathcal{N}(u, \Sigma)^{\otimes n} \parallel \mathcal{N}(v, \Sigma)^{\otimes n}) = \frac{n}{2}(u - v)^\top \Sigma^{-1}(u - v) = \frac{n}{2} \cdot \frac{4as^2}{\|\Sigma\|_{\text{op}}} = \frac{na\Delta r_s}{\sigma^2}.$$

By the Bretagnolle–Huber inequality (7, Thm. 2.2), every test ψ satisfies

$$\max_{H \in \{0,1\}} \Pr_H(\psi \neq H) \geq \frac{1}{4} \exp(-\text{KL}).$$

A procedure that sees only half-means (as DisCo-CFL does), or that does not know μ_i , has access to less information, so the bound applies to it as well. $\square$

S2.7 Proof of Proposition S4

For i in cluster k , since the class-conditionals are shared,

$$\mathcal{P}_i(y | x) = \frac{\mathcal{P}_i(y) \mathcal{P}(x | y)}{\sum_{y'} \mathcal{P}_i(y') \mathcal{P}(x | y')}, \quad q_k(y | x) = \frac{\mathcal{P}_k(y) \mathcal{P}(x | y)}{\sum_{y'} \mathcal{P}_k(y') \mathcal{P}(x | y')}.$$

Dividing, $\mathcal{P}_i(y | x) \propto q_k(y | x) \mathcal{P}_i(y) / \mathcal{P}_k(y)$, with a normalising constant that does not depend on y . Taking logarithms and the argmax gives (S4) (Lean: `prior_correction_proportional`, `prior_correction_argmax`). The Bayes classifier of i is thus a function of q_k and $\mathcal{P}_i(Y)$ only. Splitting the cluster changes neither quantity, so it cannot reduce the Bayes risk; it only reduces the data available to estimate q_k . $\square$

S2.8 Proof of Proposition S2

(i) Replacing one record by another of the same label changes one clipped summand. The half-mean therefore changes by at most $2C_{\text{clip}}/h$ in ℓ_2 (Lean: `clipped_mean_sensitivity`). The added noise has standard deviation σ_{dp} times this sensitivity, so the analytic Gaussian mechanism [5] gives (ε, δ) -DP for that vector. Every record enters exactly one released vector, so by parallel composition the whole signature is (ε, δ) -DP. The server’s computations are post-processing [8]. The class supports and half sizes are released and are not protected.

(ii) The DP noise is zero-mean and independent of everything else. The proof of Lemma S1 used only these two properties, so it still holds with

$$t_i = \frac{\text{tr } \Sigma_{\text{clip}}}{h} + p \left(\frac{2\sigma_{\text{dp}} C_{\text{clip}}}{h} \right)^2.$$

In the proof of Theorem S1, the Gaussian DP component adds a linear term with normalised standard deviation $2\sigma_{\text{dp}} C_{\text{clip}} / (h\sqrt{a})$ and a quadratic term with normalised Frobenius scale $4\sqrt{p} \sigma_{\text{dp}}^2 C_{\text{clip}}^2 / (h^2 a)$. Requiring each to be at most $\gamma / (4C_1 \sqrt{L})$ gives the two stated conditions.

(iii) This follows from the last statement of Theorem S1 with the inflated t_i . $\square$

S2.9 Proof of Proposition S3

Output clusters are conflict-free (Theorem S2(a)), so the members of A holding c share $\mathcal{P}_A^c$, and likewise for B . If $\mathcal{P}_A^c \neq \mathcal{P}_B^c$, the computation of Claim 1 with $\text{den}_{AB}^c > w_0$ gives a pooled value below τ , so $c \in \hat{D}_{AB}$. Otherwise all pairs have $\rho^c = 1$ and the pooled value is at least $(1 + \tau)/2 > \tau$ (Claim 2), so $c \notin \hat{D}_{AB}$. (Lean: `explanation_fidelity`.) $\square$

S3 Lean 4 verification

We formalised the deterministic core of every result of Section S1.2 in Lean 4 [9] with Mathlib [10]. The project is in the `lean/` directory of the repository https://github.com/vinhqdang/cluster_federated_learning and is checked with `lake build`. None of the 25 theorems uses `sorry`, and `#print axioms` reports only Lean’s standard axioms (`propext`, `Classical.choice`, `Quot.sound`) for each of them. Table S1 maps the results to Lean theorems.

The probabilistic ingredients enter the formal statements as hypotheses: sub-Gaussian and Hanson–Wright concentration, the Bretagnolle–Huber inequality and the (ε, δ) accounting of the Gaussian mechanism. For example, `claim1_conflict_not_linked` assumes the event $\mathcal{E}$ in the form $\text{num} \leq (1 - \Delta + \gamma/2) \text{den}$. These ingredients are standard, cited results that Mathlib does not yet

provide in the required form, so they are not re-proved. The combinatorial part of Theorem S2 is modelled abstractly. A state is a list of clusters and a merge step replaces two linked clusters by their union. Safety is proved for every state reachable by *any* sequence of allowed merges (`Relation.ReflTransGen`), so it does not depend on the merge order.

Table S1: Results of Section S1.2 and the Lean theorems that verify their deterministic core.

Result	Lean theorems (namespace <code>DiscoProofs</code>)
Lemma S1	<code>spearman_brown</code> , <code>population_calibration</code> , <code>raw_cosine_attenuated</code> , <code>attenuation_le_one</code>
Theorem S1 (clipping step)	<code>clip_reduces_error</code>
Theorem S2, Claims 1–2	<code>claim1_conflict_not_linked</code> , <code>claim2_compatible_linked</code> , <code>claim2_lower_bound</code>
Theorem S2(a) safety	<code>singleton_conflict_free</code> , <code>merge_preserves_conflict_free</code> , <code>safety_all_merges</code>
Theorem S2(b) maximality	<code>maximality</code>
Theorem S2(c) exact recovery	<code>exact_recovery</code>
Corollary S1	<code>sample_size_sufficient</code> , <code>evidence_condition</code>
Theorem S3	<code>lower_bound_construction</code> , <code>lower_bound_cosine</code> , <code>kl_value</code> , <code>lower_bound_sample_size</code>
Proposition S2(i)	<code>norm_clip_le</code> , <code>clipped_mean_sensitivity</code> , <code>clipped_signature_sensitivity</code>
Proposition S3	<code>explanation_fidelity</code>
Proposition S4	<code>prior_correction_proportional</code> , <code>prior_correction_argmax</code>

S4 Additional experimental details and results

S4.1 Setup

Datasets. We use four image datasets of increasing difficulty: MNIST [11], Fashion-MNIST [12] (the main benchmark), SVHN [13] (natural photographs of house numbers) and CIFAR-10 [14] (natural object images). Federations are built from them following the non-IID taxonomy used in the survey [15], which lists image classification with rotations and label swaps as the standard CFL benchmark [15, Table 6].

Heterogeneity. Each client belongs to a latent concept group. *Concept shift on features* is a rotation of the inputs (by 0° , 90° , 180° , 270° , or by $0, \theta, 2\theta, 3\theta$ in the strength sweep). *Concept shift on targets* is a label permutation that swaps two pairs of classes per group. On top of this, every client has *target skew* (Dirichlet label proportions, $\alpha = 0.3$, or $\alpha = 0.1$ in the label-only scenario) and, where indicated, *quantity skew* (log-normal dataset sizes with $\sigma = 1$, clipped to $[30, 3200]$, mean 400). Table S2 lists all scenarios. Each client has a local test set drawn with its own label proportions and transformation. We also evaluate every model on a class-balanced test set of its client’s concept (*concept accuracy*), which measures generalisation beyond the local label mix.

Training. All methods use the same CNN (two convolutional and two fully connected layers; $d = 46,730$ parameters for 28×28 inputs and 65,962 for 32×32 colour images), SGD with learning rate 0.05, batch size 32, one local epoch per round and 50 communication rounds. Methods with a warm-up (DisCo-CFL, FL+HC, FeSEM, Oracle) spend 10 of the 50 rounds on FedAvg. *Local*

Table S2: Scenarios. N : clients; groups: number of concept groups; QS: quantity skew; part.: fraction of clients per round.

Scenario	Dataset(s)	Concept groups	N	QS	Other
Rotation	FMNIST, MNIST, SVHN, CIFAR-10	4 rotations (90° steps)	40	–	
Label swap	FMNIST, MNIST, SVHN, CIFAR-10	4 labelings (2 swapped pairs)	40	–	
Rotation + QS	FMNIST	4 rotations	40	✓	
Mixed + QS	FMNIST	2 rotations $\times$ 2 labelings	40	✓	
Label skew only	FMNIST	1 (true $K = 1$)	40	✓	$\alpha = 0.1$
Minority groups	FMNIST	4 rotations, 25/9/4/2 clients	40	–	
Strength θ	FMNIST	4 rotations, $\theta \in \{15^\circ, 30^\circ, 45^\circ\}$	40	–	
Groups K	FMNIST	$K \in \{2, 4, 8\}$ labelings	40	–	
Partial participation	FMNIST	rotation / label swap	40	–	part. = 0.3
Scale	FMNIST	4 rotations	200	–	200 samples/client

trains each client alone for 50 epochs. Every configuration is run with three seeds, which change the federation, the initialisation and the sketch: 618 training runs in total, plus the clustering-only studies.

Baselines. We compare with nine methods that cover all three families of the taxonomy and the main alternatives to clustering:

- *Single model / personalisation:* FedAvg [16]; Local; FedAvg-FT (FedAvg followed by one epoch of local fine-tuning, a standard personalisation baseline).
- *Client-side:* IFCA [17].
- *Server-side:* CFL/MTCFL [18] (recursive cosine bipartitioning, triggered when the norm of the aggregated update falls below $\varepsilon_1 = 0.9$ times the mean client-update norm); FL+HC [19] (Ward clustering of local updates after the warm-up); FeSEM [20] (EM-like assignment to the nearest of K centres in parameter space, with the same warm-up and a k -means++ initialisation on the clients’ local models).
- *Metadata-based:* PACFL [21] (principal angles between client data subspaces, $p = 3$).
- *Oracle:* FedAvg inside the ground-truth groups (an upper reference).

IFCA, FL+HC, FeSEM and PACFL are *given the true number of groups K* , which favours them. DisCo-CFL and MTCFL find K on their own. The MTCFL thresholds and the FeSEM initialisation were selected on the same pilot run as DisCo-CFL. Without them, MTCFL never splits and FeSEM collapses to a single cluster.

DisCo-CFL. Sketch dimension $p = 1024$, at most 200 samples per class for the signature, classes with fewer than 4 samples not reported, $\tau = 0.8$, $\lambda = 0.5$, $e_{\min} = 0.05$, low-evidence threshold 0.3, $z = 2$, per-sample clipping $C_{\text{clip}} = 0.5$. These values, and the design choices of bottleneck linkage and evidence-ordered merging, were fixed during pilot runs on seed 0 of the Rotation, Label swap, Mixed + QS and label-only configurations (and seed 1 of Label swap with QS), using clustering quality only. They are kept identical for all scenarios, seeds and datasets. Seeds 1 and 2, the sweeps, and the MNIST, SVHN and CIFAR-10 results were not used for design. Per-sample clipping (C_{clip} chosen on seed 0 only, from $\{0.5, 1, 2\}$ in all scenarios and additionally $\{0.1, 0.25\}$ in the label-only scenario) was added after a first sensitivity study showed over-splitting caused by heavy-tailed per-sample gradients. The run without clipping is reported as an ablation. On the colour datasets we also report DisCo-T30, which uses 30 warm-up rounds (Section S4.3). The sensitivity to τ is reported in the main paper and the sensitivity to λ in Table S9.

Metrics. *Performance:* mean local test accuracy (Acc), with the prior correction of Section S1.1.4 applied identically to every method, and concept accuracy. *Fairness:* accuracy of the worst 10% of clients (Acc@10%), accuracy of the worst concept group (Worst-grp), Jain’s index. *Clustering:* ARI with respect to the groups, number of clusters K , and the *conflict-merge rate*: the fraction of conflicting client pairs (Definition S1) that end up in the same cluster. ARI penalises the legitimate placement of compatible clients (e.g. a client of a label-swap group that holds none of the swapped classes), whereas the conflict-merge rate measures exactly the harmful errors. *Statistics:* paired one-sided Wilcoxon signed-rank tests of DisCo-CFL against each baseline over all (scenario, seed) pairs.

S4.2 Full results on Fashion-MNIST

Table S3 reports all metrics on Fashion-MNIST for the ten methods. Accuracy is *with* the prior correction (S4), applied identically to every method; the uncorrected mean accuracy is shown in the first column.

Table S3: Fashion-MNIST, 40 clients, 3 seeds. Acc: mean local accuracy (with prior correction, $\pm$ std over seeds). Acc (raw): without correction. Acc@10%: mean accuracy of the worst 10% of clients. Worst-grp: accuracy of the worst concept group. Concept: accuracy on the class-balanced test set of the client’s concept. Conflict%: fraction of conflicting client pairs merged into one cluster (lower is better). IFCA, FeSEM, FL+HC and PACFL receive the true K ; DisCo-CFL finds it. Oracle uses the ground-truth groups. Best non-oracle values in bold.

Method	Acc (raw)	Acc	Acc@10%	Worst-grp	Concept	Jain	ARI	K	Conflict%
<i>Rotation</i>									
FedAvg	76.6	86.0 $\pm$ 0.5	76.5	82.9	75.7	0.99	0.00	1.0	100.0
Local	87.8	87.8 $\pm$ 0.7	80.8	86.9	60.7	1.00	0.00	40.0	0.0
FedAvg-FT	88.8	88.8 $\pm$ 0.5	81.5	88.0	77.9	1.00	0.00	40.0	0.0
IFCA	83.5	89.2 $\pm$ 1.0	80.9	86.7	77.0	1.00	0.41	4.0	22.0
FeSEM	83.0	89.1 $\pm$ 0.3	80.8	87.8	80.6	1.00	0.67	4.0	10.4
MTCFL	89.4	90.2 $\pm$ 0.6	83.6	89.1	73.9	1.00	0.15	34.7	0.0
FL+HC	78.5	86.4 $\pm$ 0.4	76.9	83.6	74.9	0.99	0.01	4.0	85.0
PACFL	82.8	89.5 $\pm$ 0.8	81.8	87.8	79.4	1.00	0.48	4.0	21.0
DisCo	84.5	90.5$\pm$0.4	84.6	89.5	83.9	1.00	1.00	4.0	0.0
Oracle	84.5	90.5 $\pm$ 0.4	84.6	89.5	83.9	1.00	1.00	4.0	0.0
<i>Label swap</i>									
FedAvg	54.2	70.1 $\pm$ 1.4	47.2	58.6	55.7	0.94	0.00	1.0	100.0
Local	88.7	88.7 $\pm$ 0.7	80.1	87.0	64.0	1.00	0.00	40.0	0.0
FedAvg-FT	90.6	90.6 $\pm$ 0.5	84.1	89.7	80.9	1.00	0.00	40.0	0.0
IFCA	76.9	85.6 $\pm$ 1.6	72.6	79.4	71.1	0.99	0.45	4.0	17.1
FeSEM	80.4	87.5 $\pm$ 5.0	76.3	83.0	79.8	0.99	0.87	4.0	5.7
MTCFL	88.2	90.4 $\pm$ 0.6	84.3	87.8	76.1	0.99	0.30	29.0	2.8
FL+HC	58.4	71.5 $\pm$ 1.1	47.0	61.1	59.1	0.93	0.01	4.0	85.7
PACFL	57.9	74.8 $\pm$ 2.6	50.7	67.8	53.6	0.94	-0.01	4.0	34.9
DisCo	85.4	90.8$\pm$0.3	84.6	89.2	84.9	1.00	1.00	4.0	0.0
Oracle	85.4	90.8 $\pm$ 0.3	84.6	89.2	84.9	1.00	1.00	4.0	0.0
<i>Rotation + QS</i>									
FedAvg	75.9	85.8 $\pm$ 0.3	78.4	82.6	74.7	1.00	0.00	1.0	100.0
Local	87.9	87.9 $\pm$ 1.2	80.4	84.9	61.9	1.00	0.00	40.0	0.0
FedAvg-FT	87.6	87.6 $\pm$ 0.7	80.8	84.8	75.4	0.99	0.00	40.0	0.0
IFCA	82.3	88.5 $\pm$ 0.6	81.7	86.1	77.0	1.00	0.44	3.3	24.3
FeSEM	81.2	88.3 $\pm$ 0.6	81.6	85.5	78.4	0.99	0.66	4.0	11.6
MTCFL	85.4	88.8 $\pm$ 0.7	82.7	86.7	73.5	1.00	0.32	20.7	15.6
FL+HC	76.9	85.3 $\pm$ 1.0	76.1	80.2	73.5	0.99	0.01	4.0	85.0
PACFL	80.1	88.0 $\pm$ 1.8	79.7	84.3	76.5	0.99	0.44	4.0	20.8
DisCo	83.3	90.1$\pm$0.2	84.5	88.7	82.1	1.00	1.00	4.0	0.0
Oracle	83.3	90.1 $\pm$ 0.2	84.5	88.7	82.1	1.00	1.00	4.0	0.0

Table S3 (continued).

Method	Acc (raw)	Acc	Acc@10%	Worst-grp	Concept	Jain	ARI	K	Conflict%
<i>Mixed + QS</i>									
FedAvg	70.2	80.8±0.9	66.3	75.1	66.6	0.98	0.00	1.0	100.0
Local	88.9	88.9±1.3	82.7	86.6	64.1	1.00	0.00	40.0	0.0
FedAvg-FT	88.6	88.6±0.9	82.9	85.7	76.5	0.99	0.00	40.0	0.0
IFCA	82.5	89.0±0.3	82.5	87.5	75.4	1.00	0.34	3.7	25.5
FeSEM	81.1	88.1±1.3	79.4	85.4	77.9	0.99	0.67	4.0	8.3
MTCFL	78.4	84.8±1.8	70.0	76.8	70.2	0.98	0.24	16.3	29.3
FL+HC	72.0	81.8±0.3	68.6	77.4	67.1	0.98	0.01	4.0	85.4
PACFL	71.5	83.3±0.4	68.3	79.3	66.4	0.98	0.20	4.0	21.8
DisCo	84.5	90.6±0.3	84.3	88.8	82.5	1.00	0.94	4.0	0.6
Oracle	84.2	90.6±0.3	84.6	88.7	83.4	1.00	1.00	4.0	0.0
<i>Label skew only</i>									
FedAvg	87.6	95.5±0.4	90.9	95.5	85.8	1.00	1.00	1.0	0.0
Local	94.6	94.6±0.2	87.7	94.6	51.0	1.00	0.00	40.0	0.0
FedAvg-FT	94.2	94.2±0.3	90.1	94.2	80.7	1.00	0.00	40.0	0.0
IFCA	87.7	95.4±0.6	90.7	95.4	85.7	1.00	1.00	1.0	0.0
FeSEM	87.7	95.5±0.4	90.6	95.5	85.7	1.00	1.00	1.0	0.0
MTCFL	92.0	95.3±0.2	88.5	95.3	77.1	1.00	0.00	20.7	0.0
FL+HC	87.5	95.5±0.5	90.4	95.5	85.9	1.00	1.00	1.0	0.0
PACFL	87.6	95.5±0.4	90.9	95.5	85.8	1.00	1.00	1.0	0.0
DisCo	87.6	95.6±0.4	90.8	95.6	85.6	1.00	0.67	1.3	0.0
Oracle	87.6	95.5±0.4	90.9	95.5	85.8	1.00	1.00	1.0	0.0
<i>Minority groups</i>									
FedAvg	77.0	85.5±0.7	74.1	35.7	76.7	0.98	0.00	1.0	100.0
Local	88.5	88.5±0.5	81.0	85.2	62.7	1.00	0.00	40.0	0.0
FedAvg-FT	89.4	89.4±0.5	82.9	81.7	78.5	1.00	0.00	40.0	0.0
IFCA	83.6	89.4±1.7	81.2	67.3	78.1	0.99	0.57	3.7	22.8
FeSEM	84.0	89.8±0.7	82.0	79.4	81.8	1.00	0.62	4.0	5.7
MTCFL	82.0	87.3±3.1	78.0	51.9	76.7	0.98	0.04	11.3	66.7
FL+HC	80.4	87.7±0.8	78.0	74.5	78.0	0.99	0.22	4.0	74.8
PACFL	83.1	89.8±0.7	83.9	82.7	78.8	1.00	0.43	4.0	17.8
DisCo	85.3	90.8±0.6	85.3	86.5	84.1	1.00	0.98	4.3	0.0
Oracle	85.3	90.9±0.6	85.3	86.5	84.2	1.00	1.00	4.0	0.0

DisCo-CFL recovers the concept structure without being told K . In *Rotation*, *Label swap* and *Rotation + QS*, DisCo-CFL finds exactly the ground-truth partition in all seeds (ARI = 1) and matches the Oracle on every metric. In *Mixed + QS*, where two groups differ only on 4 of the 10 classes, it reaches ARI 0.94 with 0.6% conflicting pairs merged. In *Minority groups* (25/9/4/2 clients) it reaches ARI 0.98 with no conflicts. The baselines that are given the true K merge 6–86% of the conflicting pairs: FeSEM 6–12%, IFCA 17–26%, PACFL 18–35% and FL+HC 75–86%. CFL/MTCFL has few conflicts only in the scenarios where it fragments the federation into 16–35 clusters. FL+HC groups clients by label proportions instead of concepts, PACFL by the pixel subspace (which a label swap leaves unchanged), IFCA suffers from initialisation, and FeSEM, the strongest clustering baseline, confuses nearby groups in parameter space.

Fairness. DisCo-CFL has the best accuracy on the worst 10% of clients in every scenario with concept shift, and the best worst-group accuracy in all of them except *Label swap*, where FedAvg-FT is 0.5 points higher (89.7% vs. 89.2%). The gap is largest for minority concepts. In *Minority groups*, the worst concept group reaches 86.5% with DisCo-CFL, against 35.7% with FedAvg, 51.9% with MTCFL, 67.3% with IFCA, 74.5% with FL+HC and 79.4% with FeSEM: the size-independence of Corollary S1 in practice. Table S4 isolates the clustering step. DisCo-CFL recovers minority groups of 1, 2, 3 and 5 clients with recall and purity 1.00 and no conflicting merges. FL+HC isolates the

minority only by spending its fixed K on it while merging 81% of the conflicting pairs among the majority groups, and PACFL does not find the minority at all.

Table S4: Minority-group recovery (clustering step only; groups of 20/10/10/ m clients; 3 seeds).

m	DisCo-CFL			FL+HC (K given)		
	recall	ARI	conflict%	recall	ARI	conflict%
1	1.00	0.98	0.0	1.00	0.09	81.5
2	1.00	0.98	0.0	1.00	0.11	81.0
3	1.00	1.00	0.0	1.00	0.12	80.6
5	1.00	0.98	0.0	0.60	0.09	82.9

S4.3 Other datasets: MNIST, SVHN and CIFAR-10

Table S6 repeats the two basic concept-shift scenarios on MNIST, SVHN and CIFAR-10 with the same hyperparameters.

MNIST and SVHN. On both datasets DisCo-CFL recovers the groups (ARI 0.97–1.00), never merges conflicting clients, and reaches the accuracy of the Oracle within 0.1 points. On SVHN, a harder dataset of natural photographs, its concept accuracy is 79.7% (rotation) and 82.5% (label swap), against at most 75.6% and 72.7% for any baseline. FeSEM is the strongest clustering baseline on these datasets (ARI 0.66–0.98), and FedAvg-FT the strongest non-clustering one. Both remain 0.7–11.6 points below DisCo-CFL in concept accuracy. In the MNIST rotation scenario, two seeds leave one or two clients in singleton clusters. These clients are flagged as anomalous (mean ARI 0.97, $K = 5.0$) and train alone; the mean accuracy still equals the Oracle’s.

CIFAR-10. CIFAR-10 is harder, and here DisCo-CFL is not the best method on every metric. With the default 10 warm-up rounds the reference model is only about 40% accurate. Its class-conditional gradients then depend little on the orientation of most natural-image classes: the mean between-group similarity is 0.83–0.90, and only vehicle classes separate clearly. The separation Δ of Assumption S2 is therefore small. For label swaps DisCo-CFL still reaches ARI 0.87 and a higher concept accuracy than every baseline (48.6%), but for rotations it reaches only ARI 0.38. The theory points to the remedy, a more informative reference model. With 30 warm-up rounds (DisCo-T30), ARI rises to 0.74 (rotation) and 0.92 (label swap), conflicting merges fall to 4.2% and 0.1%, and concept accuracy is 44.8% and 49.8%. That is the best of all non-oracle methods in both scenarios, tied with FeSEM on rotation. Table S5 gives the clustering-only trend. CFL/MTCFL has the highest *local* accuracy on CIFAR-10 (67.0% and 68.8%, above the Oracle’s 64.2% and 64.7%). It splits the federation into 33–35 clusters and personalises to each client’s label mix with the small CNN, and its concept accuracy is 7–8 points below DisCo-T30’s.

Table S5: CIFAR-10 rotation, clustering step only: effect of the warm-up length T_0 (3 seeds).

T_0	ARI	K	conflict%
10	0.37	3.7	24.8
20	0.64	5.0	8.9
30	0.74	4.7	4.2

Table S6: MNIST, SVHN and CIFAR-10 (40 clients, 3 seeds; same settings and columns as Table S3). DisCo-T30: DisCo-CFL with 30 warm-up rounds.

Method	Acc (raw)	Acc	Acc@10%	Worst-grp	Concept	Jain	ARI	K	Conflict%
<i>MNIST rotation</i>									
FedAvg	91.8	94.7±0.6	91.1	92.4	90.6	1.00	0.00	1.0	100.0
Local	96.5	96.5±0.4	94.3	96.1	82.0	1.00	0.00	40.0	0.0
FedAvg-FT	97.2	97.2±0.1	95.5	97.0	94.6	1.00	0.00	40.0	0.0
IFCA	95.5	96.9±0.5	94.4	95.6	91.2	1.00	0.42	4.0	21.6
FeSEM	97.3	98.2±0.1	97.0	97.8	95.9	1.00	0.78	4.0	5.3
MTCFL	93.8	95.7±0.7	92.8	92.9	90.8	1.00	0.09	9.7	58.9
FL+HC	91.4	94.3±0.9	88.4	91.6	89.9	1.00	0.01	4.0	85.0
PACFL	95.1	96.8±0.3	94.5	95.7	91.6	1.00	0.45	4.0	15.9
DisCo	97.6	98.5±0.1	97.3	98.3	97.1	1.00	0.97	5.0	0.0
Oracle	97.6	98.5±0.1	97.3	98.3	97.2	1.00	1.00	4.0	0.0
<i>MNIST label swap</i>									
FedAvg	60.4	78.5±2.3	52.8	61.9	60.9	0.94	0.00	1.0	100.0
Local	97.1	97.1±0.2	95.0	96.4	86.0	1.00	0.00	40.0	0.0
FedAvg-FT	98.1	98.1±0.0	96.8	97.5	92.0	1.00	0.00	40.0	0.0
IFCA	90.9	95.8±0.6	92.4	90.4	87.9	1.00	0.60	3.7	13.3
FeSEM	97.4	98.4±0.4	97.6	97.4	97.1	1.00	0.98	4.0	0.6
MTCFL	85.7	91.7±3.2	82.2	85.0	78.7	0.98	0.24	18.0	24.2
FL+HC	66.0	80.8±2.0	56.6	69.6	65.9	0.95	0.01	4.0	85.7
PACFL	67.2	83.3±2.1	62.2	75.5	63.4	0.97	0.00	4.0	30.0
DisCo	97.9	98.7±0.1	97.7	98.5	97.8	1.00	1.00	4.0	0.0
Oracle	97.9	98.7±0.1	97.7	98.5	97.8	1.00	1.00	4.0	0.0
<i>SVHN rotation</i>									
FedAvg	63.3	72.4±1.2	58.1	64.5	63.4	0.98	0.00	1.0	100.0
Local	77.5	77.5±1.0	69.0	75.6	43.9	0.99	0.00	40.0	0.0
FedAvg-FT	81.9	81.9±0.2	77.1	81.0	69.5	1.00	0.00	40.0	0.0
IFCA	74.8	80.2±1.6	71.5	76.9	65.9	0.99	0.26	3.7	25.8
FeSEM	79.8	84.1±0.3	79.0	82.3	75.6	1.00	0.66	4.0	9.4
MTCFL	82.4	84.0±0.2	79.9	83.0	66.9	1.00	0.27	24.7	10.6
FL+HC	65.3	72.9±1.4	57.6	64.9	64.4	0.98	0.01	4.0	85.0
PACFL	70.2	78.1±1.3	67.7	75.9	67.5	0.99	0.39	4.0	16.9
DisCo	81.3	85.4±0.2	80.4	83.9	79.7	1.00	0.99	4.3	0.0
DisCo-T30	80.7	84.6±0.3	80.4	83.5	78.6	1.00	0.93	5.7	0.0
Oracle	81.5	85.5±0.3	80.5	84.3	80.0	1.00	1.00	4.0	0.0

Table S6 (continued).

Method	Acc (raw)	Acc	Acc@10%	Worst-grp	Concept	Jain	ARI	K	Conflict%
<i>SVHN label swap</i>									
FedAvg	52.6	64.5±3.7	44.9	54.5	50.9	0.94	0.00	1.0	100.0
Local	81.0	81.0±1.0	75.0	79.7	50.4	1.00	0.00	40.0	0.0
FedAvg-FT	84.9	84.9±0.1	79.8	84.3	72.7	1.00	0.00	40.0	0.0
IFCA	71.4	79.0±2.6	65.2	74.9	64.1	0.98	0.37	4.0	21.2
FeSEM	73.7	80.5±6.0	71.9	76.5	70.9	0.98	0.67	4.0	12.8
MTCFL	80.0	83.1±3.4	74.0	79.4	66.5	0.99	0.26	25.0	10.7
FL+HC	56.2	66.6±3.2	45.8	57.7	53.4	0.94	0.01	4.0	84.7
PACFL	54.2	69.1±1.9	53.6	60.3	48.2	0.97	-0.01	4.0	28.9
DisCo	83.7	87.2±0.4	83.5	86.1	82.5	1.00	0.97	4.3	0.0
DisCo-T30	83.8	87.1±0.2	83.5	86.4	82.1	1.00	0.94	5.7	0.0
Oracle	83.6	87.1±0.4	83.5	86.0	82.6	1.00	1.00	4.0	0.0
<i>CIFAR-10 rotation</i>									
FedAvg	38.0	55.1±1.4	44.5	51.6	36.8	0.97	0.00	1.0	100.0
Local	56.2	56.2±3.6	42.7	52.1	21.0	0.96	0.00	40.0	0.0
FedAvg-FT	61.8	61.8±2.0	51.1	60.4	36.8	0.98	0.00	40.0	0.0
IFCA	45.9	56.5±1.6	44.1	53.4	32.0	0.97	-0.01	4.0	32.2
FeSEM	49.4	61.9±1.5	52.3	59.0	44.8	0.98	0.69	4.0	12.1
MTCFL	65.6	67.0±2.2	56.8	63.3	38.2	0.98	0.14	34.7	0.0
FL+HC	40.5	55.6±1.6	43.5	54.4	36.9	0.97	0.01	4.0	85.0
PACFL	46.2	58.6±2.4	48.1	54.8	37.3	0.98	0.37	4.0	21.2
DisCo	47.6	60.5±1.2	47.5	56.3	41.0	0.98	0.38	3.7	24.6
DisCo-T30	50.4	62.0±1.3	51.5	60.3	44.8	0.98	0.74	4.7	4.2
Oracle	52.0	64.2±1.7	54.3	61.6	47.4	0.99	1.00	4.0	0.0
<i>CIFAR-10 label swap</i>									
FedAvg	34.8	51.0±2.9	36.0	43.0	33.5	0.95	0.00	1.0	100.0
Local	56.4	56.4±3.7	42.8	51.7	21.8	0.96	0.00	40.0	0.0
FedAvg-FT	64.5	64.5±2.7	53.9	62.9	40.8	0.98	0.00	40.0	0.0
IFCA	42.9	55.0±3.5	39.8	50.7	30.7	0.95	0.08	4.0	27.0
FeSEM	46.7	59.8±3.2	46.9	55.4	43.7	0.97	0.65	4.0	11.3
MTCFL	66.9	68.8±2.2	59.0	63.7	41.4	0.98	0.19	33.3	0.5
FL+HC	38.8	53.1±2.8	35.9	48.4	34.5	0.94	0.01	4.0	85.1
PACFL	35.3	52.0±2.7	37.2	46.1	29.9	0.95	0.00	4.0	45.4
DisCo	52.5	64.1±2.4	52.3	61.5	48.6	0.98	0.87	4.3	3.0
DisCo-T30	54.7	64.6±2.2	54.9	62.4	49.8	0.99	0.92	5.0	0.1
Oracle	53.0	64.7±2.4	55.5	62.7	49.6	0.99	1.00	4.0	0.0

S4.4 Robustness sweeps

Table S7 varies one factor at a time on Fashion-MNIST: the strength of the concept shift, the number of groups, the fraction of participating clients and the number of clients. Baselines are given the true K where they need it.

Table S7: Robustness sweeps (Fashion-MNIST, 3 seeds). For DisCo-CFL: ARI, number of clusters, conflict-merge rate, accuracy and concept accuracy. For comparison: the best baseline on accuracy and on concept accuracy (in parentheses), and the Oracle’s accuracy.

Factor	Setting	DisCo-CFL					Best baseline Acc	Best baseline Concept	Oracle Acc
		ARI	K	Conflict%	Acc	Concept			
Strength	$\theta = 15^\circ$	0.42	3.0	32.1	90.0	82.8	90.2 (FeSEM)	83.0 (FeSEM)	90.5
Strength	$\theta = 30^\circ$	0.93	4.3	1.7	90.3	83.1	90.0 (PACFL)	81.2 (PACFL)	90.4
Strength	$\theta = 45^\circ$	1.00	4.0	0.0	90.4	83.8	89.9 (FeSEM)	81.8 (FeSEM)	90.4
Strength	$\theta = 90^\circ$	1.00	4.0	0.0	90.5	83.9	90.2 (MTCFL)	80.6 (FeSEM)	90.5
Groups	$K = 2$	0.94	3.3	0.0	91.4	86.5	88.5 (IFCA)	80.2 (IFCA)	91.5
Groups	$K = 4$	1.00	4.0	0.0	90.8	84.9	90.6 (FedAvg-FT)	80.9 (FedAvg-FT)	90.8
Groups	$K = 8$	0.92	8.0	0.3	90.4	81.7	88.0 (FeSEM)	75.4 (FeSEM)	90.5
Participation 30%	rotation	1.00	4.0	0.0	89.1	78.4	87.5 (PACFL)	75.1 (FeSEM)	89.1
Participation 30%	label swap	0.98	4.0	0.0	89.3	78.3	86.5 (FeSEM)	73.0 (FeSEM)	89.3
Scale	$N = 200$	0.98	6.3	0.0	91.1	86.6	89.8 (PACFL)	83.9 (FeSEM)	91.1

Strength of the shift. With rotations of $0, \theta, 2\theta, 3\theta$ degrees, DisCo-CFL separates the groups perfectly from $\theta = 45^\circ$ on and almost perfectly at 30° (ARI 0.93). At 15° neighbouring concepts are so close that their class-conditional gradients agree above the threshold, and DisCo-CFL merges them (ARI 0.42). This is the behaviour predicted by the separation assumption: the method groups what it cannot tell apart. It costs little, because nearby concepts can share a model: accuracy is 90.0% against 90.5% for the Oracle, and 0.2 points below FeSEM, which is given $K = 4$. For every $\theta \geq 30^\circ$, DisCo-CFL has the best accuracy and concept accuracy and is within 0.1 points of the Oracle.

Number of groups. With $K = 2, 4$ and 8 label-permutation groups, DisCo-CFL finds K without being told (3.3, 4.0 and 8.0 clusters; the extra cluster at $K = 2$ is a conflict-free split) and merges at most 0.3% of conflicting pairs. Its accuracy is within 0.1 points of the Oracle for all K . The margin over the best baseline is 2.9, 0.2 and 2.4 points in accuracy and 6.3, 4.0 and 6.3 points in concept accuracy for $K = 2, 4, 8$. With $K = 2$ and $K = 8$ the baselines given K confuse groups (IFCA and FeSEM merge 25% and 4% of the conflicting pairs), whereas with $K = 4$ the fine-tuned global model is a strong competitor on local accuracy only.

Partial participation. When only 30% of the clients take part in each round, the warm-up model is noisier. DisCo-CFL still recovers the groups (ARI 1.00 and 0.98) and matches the Oracle, while the best baseline is 1.6–2.8 points below in accuracy and 3.3–5.3 points below in concept accuracy.

Number of clients. With 200 clients of 200 samples each, DisCo-CFL reaches ARI 0.98 with no conflicting merge. The extra clusters ($K = 5$ –7) are conflict-free splits of groups; three of them are singletons flagged as anomalous in the audit log. Accuracy equals the Oracle’s (91.1%), 1.3 points above the best baseline, and concept accuracy is 2.7 points above the best baseline.

S4.5 Statistical significance

Table S8 compares DisCo-CFL with each baseline over all (scenario, seed) pairs with concept shift that both methods were run on. DisCo-CFL is significantly better than every baseline on mean accuracy, concept accuracy and worst-group accuracy (one-sided Wilcoxon signed-rank tests, all $p < 0.01$). The closest competitors are FeSEM, MTCFL and FedAvg-FT. MTCFL and FedAvg-FT win on local accuracy mainly on CIFAR-10 (5–6 of the 6 CIFAR-10 pairs), and FeSEM wins on 8 of 57 pairs, mostly CIFAR-10 rotation and the 15° rotation. On concept accuracy DisCo-CFL wins against every baseline in at least 84% of the pairs.

Table S8: Paired one-sided Wilcoxon signed-rank tests of DisCo-CFL against each baseline over all (scenario, seed) pairs with concept shift. W/T/L: wins, ties ($|\Delta| < 0.05$ points) and losses of DisCo-CFL.

Baseline	pairs	Accuracy		Concept accuracy		Worst-group accuracy	
		W/T/L	p	W/T/L	p	W/T/L	p
FedAvg	57	57/0/0	2.6e-11	57/0/0	2.6e-11	57/0/0	2.6e-11
Local	33	33/0/0	1.2e-10	33/0/0	1.2e-10	32/0/1	2.3e-10
FedAvg-FT	33	27/1/5	2.3e-5	33/0/0	1.2e-10	24/0/9	0.004
IFCA	54	54/0/0	8.1e-11	54/0/0	8.1e-11	54/0/0	8.1e-11
FeSEM	57	45/4/8	2.5e-8	48/2/7	1.4e-8	47/3/7	5.2e-8
MTCFL	39	30/2/7	0.002	39/0/0	1.8e-12	30/0/9	1.4e-4
FL+HC	57	57/0/0	2.6e-11	57/0/0	2.6e-11	57/0/0	2.6e-11
PACFL	57	54/0/3	7.0e-11	57/0/0	2.6e-11	53/1/3	5.1e-11

S4.6 Calibration across sample sizes and shrinkage

Calibration across sample sizes. Varying the client size from 50 to 800 samples (Figure S1), the mean disattenuated similarity between clients of the same group stays at 0.98–0.99, whereas the raw cosine stays around 0.91–0.95 and depends on the per-class sample counts. ARI reaches 1.00 from 200 samples per client on (rotation) and from 400 (label swap). Table S9 shows that the shrinkage λ has almost no effect between 0.02 and 1, including values that satisfy the condition of Corollary S1. Figure S2 shows, on synthetic signatures, that recovery of a minority group does not depend on its size.

Table S9: Sensitivity to the shrinkage λ at $\tau = 0.8$: ARI / number of clusters (3 seeds, clustering step). For the label-only scenario the correct K is 1.

Scenario	$\lambda = 0.02$	$\lambda = 0.1$	$\lambda = 0.2$	$\lambda = 0.5$	$\lambda = 1.0$
Rotation	1.00 / 4.0	1.00 / 4.0	1.00 / 4.0	1.00 / 4.0	0.98 / 4.0
Label swap	1.00 / 4.0	1.00 / 4.0	1.00 / 4.0	1.00 / 4.0	1.00 / 4.0
Rotation + QS	0.99 / 4.3	0.99 / 4.3	0.99 / 4.3	1.00 / 4.0	0.98 / 4.0
Mixed + QS	0.96 / 4.0	0.94 / 4.0	0.94 / 4.0	0.94 / 4.0	0.91 / 4.0
Label skew only	0.67 / 1.3	0.67 / 1.3	0.67 / 1.3	0.67 / 1.3	0.67 / 1.3
Minority groups	0.98 / 4.3	0.98 / 4.3	0.98 / 4.3	0.98 / 4.3	1.00 / 4.0

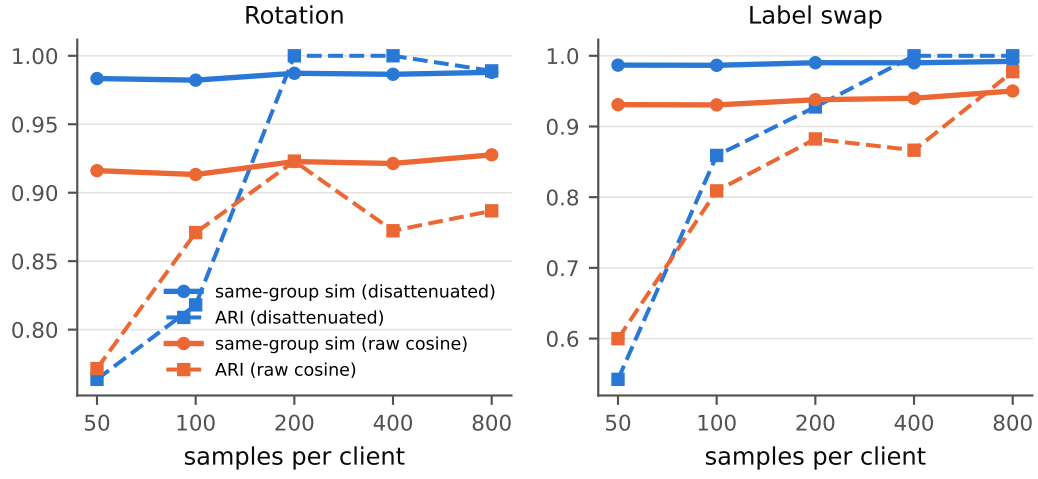


Figure S1: Same-group similarity (solid) and ARI (dashed) as a function of the client size. Blue: disattenuated; orange: raw cosine.

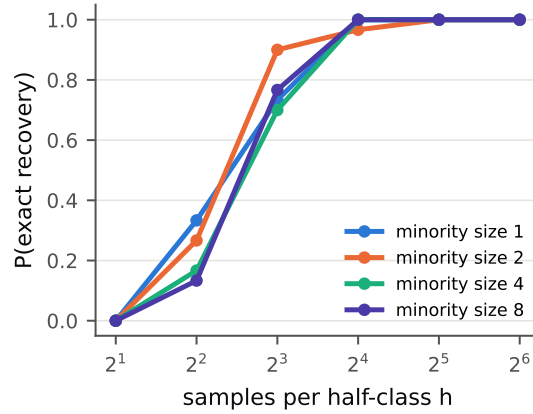


Figure S2: Synthetic validation of Corollary S1: exact recovery of a minority group does not depend on its size.

S4.7 Cost and scalability

Each client uploads its signature *once*, about 1.2×10^4 floats with $p = 1024$. The size is independent of the model dimension d ; IFCA, by contrast, downloads K models per client in *every* round, and CFL/FL+HC cluster d -dimensional updates. Computing the signature takes 0.14–0.16 s per client on one CPU core. Server-side clustering takes 0.05 s for 40 clients and 2.3 s for 320 clients, with $\text{ARI} \geq 0.98$ at every scale.

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